

## **Machine Learning Scenario Based Answer 2**

**1. A bank wants to predict whether a loan applicant will default based on credit score, income, and past loan history. What type of ML problem is this, and what steps would you take to solve it?**

### **Loan Default Prediction**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect applicant details.
2. Clean and preprocess data.
3. Select features (credit score, income, loan history).
4. Split data into train and test sets.
5. Train a classification model.
6. Predict whether the customer will default.
7. Evaluate accuracy.
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**2. A retail store wants to predict the demand for different products to optimize inventory levels. What type of ML problem is this, and what steps would you take to solve it?**

### **Product Demand Prediction**

**Problem Type:** Supervised Learning – Regression

**Logic:**

1. Collect past sales data.
2. Clean and preprocess data.
3. Select important features.
4. Train a regression model.
5. Predict future demand.
6. Use predictions for inventory planning.

**3. A factory wants to detect whether a manufactured product is defective based on sensor readings and quality control data. What type of ML problem is this, and what steps would you take to solve it?**

### **Defective Product Detection**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect sensor and quality data.
2. Label products as defective or non-defective.
3. Preprocess data.
4. Train classification model.
5. Predict product quality.
6. Evaluate performance.

**4. A healthcare provider wants to analyze patient symptoms and classify them into different disease categories. What type of ML problem is this, and what steps would you take to solve it?**

### **Disease Classification**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect patient symptoms.
2. Clean medical data.
3. Train classification model.
4. Predict disease category.
5. Evaluate accuracy.

**5. An e-commerce company wants to identify and remove fake reviews posted by bots or fraudsters. What type of ML problem is this, and what steps would you take to solve it?**

### **Fake Review Detection**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect reviews and user details.
2. Label reviews as genuine or fake.
3. Convert text into numerical features.
4. Train classification model.
5. Detect fake reviews.
6. Remove suspicious reviews.

**6. A financial firm wants to predict stock price movements based on historical price data and market indicators. What type of ML problem is this, and what steps would you take to solve it?**

### **Stock Price Movement Prediction**

**Problem Type:** Supervised Learning – Regression

**Logic:**

1. Collect historical stock data.
2. Preprocess data.
3. Select market indicators.
4. Train regression model.
5. Predict future stock prices.
6. Evaluate using  $R^2$  Score.

**7. A social media platform wants to detect fake user accounts based on user activity and profile data. What type of ML problem is this, and what steps would you take to solve it?**

### **Fake User Account Detection**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect user activity and profile data.
2. Label accounts as genuine or fake.
3. Train classification model.
4. Predict fake accounts.
5. Take action on suspicious accounts.

**8. A marketing agency wants to segment customers into different groups based on their purchasing behavior. What type of ML problem is this, and what steps would you take to solve it?**

### **Customer Segmentation**

**Problem Type:** Unsupervised Learning – Clustering

**Logic:**

1. Collect purchasing behavior data.
2. Clean and preprocess data.
3. Apply clustering algorithm.
4. Group similar customers.
5. Use groups for targeted marketing.

**9. A geospatial research team wants to analyze satellite images to classify different land types (forest, water, urban). What type of ML problem is this, and what steps would you take to solve it?**

### **Land Type Classification**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect satellite images.
2. Label images (forest, water, urban).
3. Preprocess images.
4. Train image classification model.
5. Predict land type.
6. Evaluate model performance.

**10. A streaming service wants to predict which users are likely to cancel their subscriptions. What type of ML problem is this, and what steps would you take to solve it?**

### **Subscription Cancellation Prediction**

**Problem Type:** Supervised Learning – Classification

**Logic:**

1. Collect user activity data.
2. Label users who cancelled and who stayed.
3. Preprocess data.
4. Train classification model.
5. Predict likely cancellations.
6. Take retention actions.